KAHM as an optional forecasting component

A fresh policy augmentation study with a matched search control
Research report | 1 November 2026

The prespecified policy advantage is not established

Abstract

We evaluate whether adding a bounded, centered in-sample KAHM residual correction improves a strong validation-selected forecasting policy. P0 is the established non-KAHM menu; P1 adds the KAHM correction; P0+ adds a raw-history Nyström RBF residual family with the same number of additional validation scores. All policies retain training-only feature construction, B-selected preprocessing and dimensions, and C-selected calibration and model family. On 24 fresh simulation repetitions, P1 has 0.82% lower nonlinear standardized MSE than P0 and 0.66% lower than P0+. The study establishes 0 of the two prespecified 3% superiority criteria and 2 of the two 5% linear noninferiority criteria. On 48 M4 series outside the recorded prior pilot, from two frequency panels, policy loss is 2.27% higher than P0; KAHM is selected for 3 series. Natural-series findings are descriptive benchmark-development evidence. They are analyzed separately from the simulations.

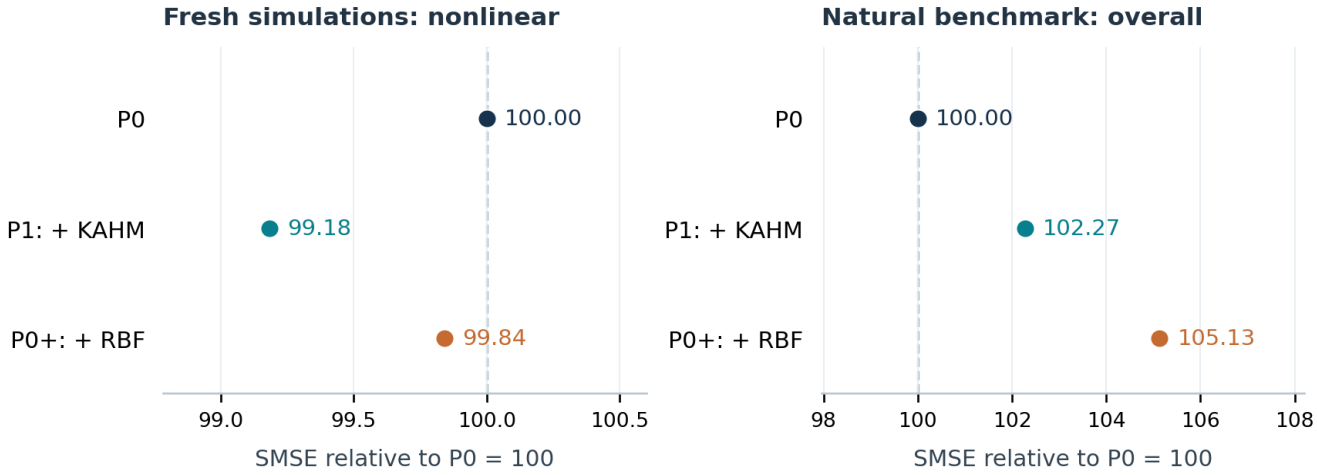


Figure 1. Calibrated policy losses, each panel indexed to its own P0. Lower values indicate lower loss. Points are descriptive; the four adjusted synthetic decisions appear in Table 3. The panels use different evaluation designs and are not pooled.

The experiment tests incremental value at the level of a complete selection procedure. Improvement over one shared common forecast remains a separate, conditional mechanism result. The presence of an optional KAHM candidate does not guarantee a smaller terminal loss.

1. Policies, candidate selection and data roles

The scientific question concerns a deployable choice made before future outcomes are known. P0 is selected from credible non-KAHM candidates. The two augmented menus differ by one additional candidate family, so their comparison addresses the value of KAHM beyond an equally sized increase in validation search opportunities. Selection itself can overfit a finite development sample [3].

Policy	Additional family	Simulation / natural family count
P0	Established non-KAHM candidates	10 / 11
P1	P0 + centered in-sample KAHM residual correction	11 / 12
P0+	P0 + bounded raw-history Nyström RBF residual correction	11 / 12

P0 includes ridge, raw Nyström RBF, histogram gradient boosting, persistence, common geometry, bounded and unbounded local-RBF departures, common plus shared bias, and in-sample local-RBF correction. The simulation menu also includes an episode cross-fitted RBF correction. The natural menu omits that family and adds seasonal naive and drift forecasts; no independent episode folds are invented from a single observed series.

Role	Information used	Fresh simulations	Natural series
A: fit	Feature maps, transformations, coefficients; fixed loss scale	64 episodes; 512 rows	256-512 recent windows; realized minimum 291
B: tune	Penalties, cap/scaler; one shared residual bias	32 episodes; 256 rows	64 origins
C: choose	Correction multiplier and one complete family	32 episodes; 256 rows	64 origins
D: evaluate	Frozen predictions, then released targets	128 episodes; 1,664 rows	32 rolling origins

Chronology and information available at a forecast origin

All horizons are {1, 4, 12} and the tabular history length is 12. Hyperparameters and coefficients are not refitted after C. In the natural panel, twelve intervening origins ensure that all targets from one block are mature before the next block's first origin; target observations are disjoint between blocks. Later D contexts may include earlier D outcomes already observed at that origin [4].

The simulation cohort uses independent episode splits, with 13 retained candidate origins per episode before row caps. The natural design uses one continuous record per series with overlapping histories within each block. These different sources of dependence are respected in their different analysis rules.

2. Residual construction and complete tuning grids

A shared baseline and a centered correction

The common predictor g is fitted on all A rows and tuned on B. A single shared bias is the B mean of y - g(x). The correction regresses the in-sample A residuals on bounded features centered at their A mean. Its response-mean intercept is omitted. The forecast is g(x) + b + eta times the centered feature slope; eta = 0 recovers g(x) + b exactly.

Component	Frozen settings and search	Selection role
History transforms	Standard, min-max, tanh(z); tanh multiplier 1	Fit on A; choose on B
New local corrections	Native caps {1, 2, 4, 8, 12}; each of the three transforms	One cap/scaler across horizons
Legacy geometry	Caps {1, 2, 4, 8, 12, 20}; common/departure transform pairs (s,s), (m,m), (t,t), (t,m), (m,t)	B; cap 20 duplicates 12
Ridge penalties	{0.0001, 0.001, 0.01, 0.1, 1}	B, separately per horizon
Calibration	{0, 0.25, 0.5, 1}; legacy blends toward ridge, new residual families toward common + bias	C, separately per horizon
Raw RBF	64 landmarks; bandwidth factors {0.5, 1, 2}; three history transforms	B; inherited complete head
Extra raw-RBF residual	64 landmarks; bandwidth factors {0.25, 0.5, 1, 2, 4}; three history transforms	B; one new family on C
Gradient boosting	Learning rates {0.05, 0.1}; leaves {7, 15}; 100 iterations; L2 penalty 1	B; per-horizon candidate
Geometry reference bank	Partitions {2, 4, 8}; at most 24 references per local model; bank fit cap 512 rows	A only; settings fixed

Table 1. Complete selected grids for the inherited comparison and the new residual families. s, m and t denote standard, min-max and tanh. Persistence, seasonal naive and drift have no fitted search grid; the two direct natural controls are specified in Section 5.

For every correction map, B chooses the ridge penalty separately per horizon and then chooses one map using the mean normalized loss across horizons. KAHM and the extra raw-RBF family each receive 15 feature configurations and 75 B scores per horizon, followed by one extra C family slot. Family selection uses calibrated C loss. Ties prefer the first candidate in the frozen order, with existing P0 candidates preceding an added family.

Min-max queries are not clipped. Tanh is applied to A-standardized histories with multiplier one and no subsequent history standardization. Post-map feature scaling and radial bounding are separate operations. Bandwidth factors multiply the median squared landmark distance. The feature scale floor is 10^{-6}, the target-variance floor is 10^{-12}, and the inherited relative normalizer guard is 10^{-10}.

3. Fresh simulations: the four primary criteria

The cohort comprises linear AR, nonlinear AR, Van der Pol and Duffing laws, each at added-noise levels 0 and 0.1, with 24 new paired repetitions: 192 tasks. This is a new cohort under the previously studied laws, with master seed 2026110191. The independent inferential unit is the repetition; origins, horizons and task cells within a repetition are not additional replications.

Policy	Nonlinear mean SMSE	Linear mean SMSE
P0	0.027268	0.913907
P1	0.027046	0.913922
P0+	0.027225	0.913937

Table 2. Horizon losses are divided by the corresponding A-target population variance, floored at 10^{-12}. Horizons, noise levels and laws are averaged equally within each repetition; reported values then average the 24 repetition summaries. Nonlinear and linear panels are separate.

Comparison	Required margin	Point gain	Adjusted upper bound	Decision
Nonlinear: P1 / P0	3% superiority	+0.82%	0.000711	Not established
Nonlinear: P1 / P0+	3% superiority	+0.66%	0.000798	Not established
Linear: P1 / P0	5% noninferiority	-0.0016%	-0.044648	Established
Linear: P1 / P0+	5% noninferiority	+0.0016%	-0.044677	Established

Table 3. Positive point gains mean lower P1 loss. The upper bound concerns the margin-adjusted loss difference, not the percentage gain. Every frozen criterion is reported.

Decision rule

For a nonlinear comparison, define the repetition difference as P1 loss minus 0.97 times comparator loss. For linear noninferiority use P1 loss minus 1.05 times comparator loss. The reported endpoint is the mean difference plus the 98.75th percentile of a Student-t distribution with 23 degrees of freedom times its standard error. A criterion is established only if this endpoint is strictly negative. The common critical value is 2.397875. Bonferroni adjustment covers exactly the four prespecified claims.

2 of 4 criteria established: 0/2 nonlinear superiority criteria and 2/2 linear noninferiority criteria.

Failure to establish a 3% advantage does not demonstrate equality or rule out a smaller effect. Passing linear noninferiority addresses its declared loss margin and does not establish nonlinear superiority. The Student-t calculations are approximate finite-repetition inference under the stated simulation population; they are not distribution-free certificates.

4. Synthetic heterogeneity and optional selection

Condition summaries describe how the complete policies behave across the fixed laws and noise levels. They are not additional confirmatory tests and are not used to change the menus, dimension grid, calibration rule or primary interpretation.

Law / noise	P0	P1	P0+	P1 gain vs P0	KAHM chosen
Linear AR noise 0	0.91259	0.91259	0.91247	+0.00%	0/24
Linear AR noise 0.1	0.91522	0.91525	0.91541	-0.0032%	1/24
Nonlinear AR noise 0	0.01945	0.01946	0.01945	-0.06%	5/24
Nonlinear AR noise 0.1	0.03168	0.03169	0.03168	-0.01%	6/24
Van der Pol noise 0	0.00410	0.00401	0.00410	+2.05%	2/24
Van der Pol noise 0.1	0.05389	0.05296	0.05365	+1.73%	14/24
Duffing noise 0	0.01102	0.01087	0.01101	+1.34%	11/24
Duffing noise 0.1	0.04347	0.04328	0.04346	+0.44%	5/24

Table 4. Mean SMSE in each law/noise cell, averaged over horizons and 24 repetitions. KAHM choice counts refer to C selection by P1. The same task can have distinct cap, scaler and penalty choices for its KAHM and non-KAHM candidates.

What changes when KAHM is selected?

Nonlinear panel. KAHM is selected in 43/144 tasks. Within that C-defined subset, P1 wins against P0 on 30 terminal task losses, ties on 0, and loses on 13. Its mean loss is 2.30% lower than P0. The subset is descriptive and cannot replace the full-policy criterion.

Linear panel. KAHM is selected in 1/48 tasks. Within that C-defined subset, P1 wins against P0 on 0 terminal task losses, ties on 0, and loses on 1. Its mean loss is 0.08% higher than P0. The subset is descriptive and cannot replace the full-policy criterion.

As a standalone correction in the nonlinear panel, in-sample KAHM has 12.50% lower SMSE than its shared common-plus-bias baseline and 3.83% lower than the independently selected local-RBF correction. These are descriptive mechanism comparisons. A useful correction to a fixed baseline need not improve a stronger policy that can choose among different complete forecasts.

5. Natural series: external benchmark development

The observed-data panel contains 24 weekly and 24 hourly series selected by a fixed identity hash and length eligibility from the pinned M4 Train files [2]. Seventy-two previously used M4 IDs were excluded. Official M4 Test files were not downloaded. Each series contributes 32 rolling origins at horizons {1, 4, 12}; this is not the official M4 evaluation. Anonymous series can share underlying sources, so no independent-series confidence interval or population-level significance claim is made.

Policy	Weekly SMSE	Hourly SMSE	Overall SMSE	Frequency MASE
P0	0.07939	0.17583	0.12761	1.3809
P1	0.08392	0.17710	0.13051	1.3902
P0+	0.09146	0.17686	0.13416	1.4049

Table 5. Each frequency mean weights its 24 series equally; the overall mean weights the two frequencies equally. Frequency MASE uses A-only mean absolute differences at lag 1 for weekly and 24 for hourly series. Seasonal MASE, reported in the metrics record, uses lags 52 and 24. Denominators are floored at 10^{-12} and are never fitted on D.

P1 has 2.27% higher overall SMSE than P0 and 2.72% lower than P0+. KAHM is selected for 3/48 series. The remaining P1 forecasts use the selected P0 family with unchanged parameters.

Selected series	Cap	Scaler	P0 SMSE	P1 SMSE	P1 gain
W160 Weekly	1	Tanh	0.00306	0.00264	+13.67%
W196 Weekly	4	Standard	0.54429	0.65350	-20.06%
H368 Hourly	2	Tanh	0.05553	0.08595	-54.77%

Table 6. Every natural-series case in which P1 selects KAHM. These outcomes describe a subset chosen on C; D losses do not define the subset or revise the selection rule.

Conditional correction value and policy value

Across all 48 series, standalone KAHM correction has 3.33% lower SMSE than the identical common-plus-bias baseline and 2.15% lower than local-RBF correction. The policy comparison asks a broader question because P0 may instead select a raw method, persistence, drift or seasonal naive. The fixed-baseline result therefore does not establish that adding KAHM improves the available forecasting policy.

Seasonal naive accesses its declared lag from the observed history; drift uses the earliest observation in A's retained history interval and the current observation. All forecasting parameters remain fixed after C, while each origin receives the observations then available. The short terminal period, two frequency panels and fixed finite menus limit external scope.

6. Validation-selected dimensions and preprocessing

The experiment carries forward the corrected dimension policy and earlier validation-based dimension studies. A requested cap c initializes each local model at min(c, 12, N_c-1); the inherited positive-variance and projected-range rules can reduce the realized rank. The native default cap 20 is equivalent to cap 12 at ambient history dimension 12. The legacy search retains this duplicate and resolves ties toward 12; the new corrections search the five distinct caps.

Cohort / head	1	2	4	8	12	Std	Min-max	Tanh
Simulation Common	42	29	60	41	20	79	65	48
Simulation KAHM	44	71	45	24	8	65	93	34
Simulation Local RBF	76	92	20	3	1	71	73	48
Natural Common	12	7	4	7	18	13	16	19
Natural KAHM	12	10	15	5	6	12	16	20
Natural Local RBF	22	7	9	3	7	10	14	24

Table 7. B-selected requested caps and transforms across all tasks: 192 simulation tasks and 48 natural tasks per head. These are candidate settings even where P1 ultimately chooses a different complete family. They are not a ranking of transforms on D.

Realized local ranks

Simulation KAHM. 2,688 local-model occurrences in selected banks; realized rank 0-12, mean 3.39. Counts by realized rank (rank: occurrences): 0: 5, 1: 617, 2: 994, 4: 627, 8: 335, 10: 2, 12: 108. Zero-rank fallbacks: 5.

Natural KAHM. 672 local-model occurrences in selected banks; realized rank 1-12, mean 4.25. Counts by realized rank (rank: occurrences): 1: 168, 2: 140, 4: 210, 8: 70, 12: 84. Zero-rank fallbacks: 0.

Cohort	KAHM features	Local RBF features	Extra raw-RBF features
Simulation	168-168 mean 168.0	168-168 mean 168.0	66-76 mean 75.8
Natural	168-168 mean 168.0	168-168 mean 168.0	59-76 mean 75.0

Table 8. Widths of the selected correction designs. The extra family includes raw history plus up to 64 Nyström directions, followed by feature scaling and radial bounding. Actual width and effective rank depend on the fitted data.

No transform receives a presumption of superiority. The cap and scaler summaries describe the frozen B-selection procedure; terminal D loss never selects a dimension.

7. Computation, verification and limitations

Measurement	Simulations (192 tasks)	Natural (48 tasks)
Fit worker wall time, summed	2,723.65 seconds	382.68 seconds
Fit worker CPU time, summed	2,718.92 seconds	381.13 seconds
Prediction worker wall time, summed	269.58 seconds	10.94 seconds
Extra-RBF fit wall time, summed	17.72 seconds	4.20 seconds
Largest individual worker RSS	301.09 MiB	238.59 MiB
Complete compressed fitting evidence	1,808.99 MiB	186.87 MiB

Table 9. Durations sum individual worker measurements; concurrent execution means these are not elapsed stage times. Peak worker RSS includes interpreter and retained process state. The compressed fitting evidence stores candidates, diagnostics and audit material and is not a minimal deployed model size. All costs include the declared rejected searches and diagnostic procedures appropriate to their recorded scope.

Independent saved-record verification

Both independent audits passed. The simulation audit covers 192 tasks and 9,600 retained forecast matrices; the natural audit covers 48 series and 2,016 matrices. Recorded hashes link source/configuration freezes, fitted states, prediction seals and released targets. The audits reconstruct loss denominators, B and C choices, complete policy forwarding and reported aggregates. Manufactured tests independently checked ridge normal equations, zero-correction recovery, query-batch invariance and exact in-sample adapter parity.

The inherited numerical policy first attempts the original ridge SVD and invokes a logged alternative driver only on a linear-algebra exception. Simulation fitting recorded 1 invocation(s), 1 recovered and 0 unrecovered; natural fitting recorded 0, 0 and 0, respectively. Requested dimensions, realized ranks and native fallbacks are retained separately from these numerical-solver events.

Limits of the evidence

The design was informed by earlier studies and was fixed before this cohort's outcomes; it is not a public preregistration. Fresh simulation draws extend evidence within four known laws. They do not establish performance over a population of real forecasting tasks. Natural series were selected from available benchmark metadata, and unknown dependencies between series prevent treating 48 cases as 48 independent source populations.

Equal added score counts do not match computation, feature width or statistical capacity. The experimental suite shares fitted objects, so total timings do not identify every isolated policy's marginal runtime. The ridge correction has a deterministic magnitude bound, but that bound does not certify lower forecasting risk. Procedural target masking and saved-array checks do not constitute cryptographic isolation or a complete independent reimplementation of every feature map.

8. Scientific interpretation and reproducibility

The strongest defensible position remains a geometry-dependent nonlinear feature family with conditional value as a bounded, calibrated residual correction. The new experiment does not establish the prespecified incremental advantage of adding that correction to a strong forecasting policy. Fixed-common improvements and stable correction magnitudes remain narrower findings with different estimands.

The distinction matters for the manuscript's central argument. A representation may add nonconstant directions beyond a shared coordinate basis, yet the available alternatives and finite-sample selection error determine whether those directions improve the forecast actually chosen. Retaining unsuccessful policy comparisons clarifies that boundary and supports a more precise scientific contribution.

A justified continuation is a declared exploratory analysis of selection stability and residual geometry using these completed records. Any diagnostic or revised acceptance rule developed from this evidence requires another reserved evaluation. The current D outcomes must not be reused as untouched confirmation for a policy designed after inspecting them.

Reproducibility record

The simulation source was published before terminal D generation. The natural source was frozen locally before numerical parsing and published after its evaluation. The combined source commit below contains both study runners and configurations; it is not a public preregistration. The retained extensions contain the protocols, grids, source manifests, sealed predictions, audit programs and analysis tables. Reporting reads verified metrics without refitting a model or changing a decision.

```
Branch: research/kahm-policy-augmentation-2026-11-01

Combined source commit: b1915813e98a59572a4bbb800adc2bd1f3fccc1b

M4 input snapshot commit: 6c1067e5a57161249b17289a565178dc7a3fb3ca

Extensions: policy_augmentation_20261101 and
policy_augmentation_natural_20261101
```

References and data provenance

[1] KAHM forecasting research repository. Frozen scientific source and study protocols. Pinned study source.

[2] M4 Competition. M4-methods repository, pinned Train data and metadata snapshot. M4 source snapshot.

[3] Cawley, G. C. and Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research, 11, 2079-2107. Article and paper.

[4] Hyndman, R. J. and Athanasopoulos, G. Forecasting: Principles and Practice, third edition. Chapter 5: The forecaster's toolbox; simple forecasting methods, accuracy evaluation and time-series cross-validation. Open textbook.

Evidence source: KAHM_Policy_Augmentation_Metrics.json. Every numerical statement in this report is drawn from the verified record or from explicitly labeled arithmetic on its tables. Complete input hashes are retained with that record.